

Synthetic Vision: Recovering Latent Geometry in Architectural Heritage Fragments

Clay SEIFERT*, Patrick DANAHY^a

*University of Texas at Austin
310 Inner Campus Drive, Austin, TX 78712
clayhseifert@gmail.com

^a Georgia Institute of Technology

Abstract

We present a two-stage computational framework for analyzing the underlying projective geometry of architectural fragments. Our approach first trains a Vision Transformer (ViT) on a synthetic dataset of modeled geometric primitives, allowing the network to learn key generative operations common in architectural form. This pretrained model is then adapted to localize these geometric cues in real fragments captured from four historic buildings using photogrammetry and orthographic projection. By segmenting and embedding these saliency maps, we reveal how regions of real fragments correspond to abstract geometric lineages, enabling direct comparison and systematic cataloging across synthetic and real domains. This demonstrates that—even when trained on purely synthetic data—a ViT-based patch-probe can generalize to photogrammetric fragments to recover latent projective structure, thereby offering a scalable and objective method to assist in architectural heritage preservation, restoration accuracy, and digital reconstruction practices.

Keywords: patch-probe saliency, latent geometry, projective transformations, photogrammetry fragments, Vision Transformer, UMAP embedding

1. Introduction

Architectural fragments—remnants of vaults, capitals, and moldings—frequently constitute the only surviving physical evidence of lost buildings or stylistic systems. In photogrammetric documentation, these fragments are typically captured as high-resolution 3D meshes. While it is possible to extract orthographic projections or clipped sub-fragments from these models, identifying and quantifying underlying projective geometry remains largely a manual, expertise-intensive task. Determining, for example, whether a broken curvature reflects a true semicircular arch, a segmental arch, or a partial cylinder, requires significant geometric interpretation.

Recovering latent geometry is essential: geometric structure encodes original construction methods, supports structural analysis, and is a prerequisite for “in kind” repair or digital reconstruction that preserves historical integrity. If generative geometry is not understood, interventions risk introducing incompatible technical or aesthetic vocabularies.

Most existing computational approaches in fragment analysis rely on direct 3D reconstruction (e.g., point-cloud meshing or NURBS fitting) or on semantic classification of

fragments (e.g., cornice, vault, rib). These methods generally treat geometry as an outcome rather than as an explicit generative process, and consequently do not recover the sequences of projective operations (e.g., circle-to-ellipse, line-to-conic, cylinder-to-half-cylinder) that are central to architectural analysis and restoration.

This paper introduces a two-stage vision pipeline to recover such latent geometric cues directly from 2D image data:

1. Stage 1: Synthetic Pretraining. A dataset of 56 synthetic geometry classes is constructed, each defined by a reproducible sequence of primitive operations (e.g., CIRCLE → EXTRUDE → BOOLSUB_RECTANGLE). Each class is rendered under 38 isometric camera angles and 8 lighting/ground conditions (304 images per class; 17,024 total). A Vision Transformer (ViT) is trained to classify these 56 lineage classes from 2D projections, compelling the network to encode projective transformations as visual features.
2. Stage 2: Fragment Analysis via Patch-Probe. The ViT backbone is frozen, and a lightweight patch-probe MLP head is attached, producing a 14×14 grid of per-patch logits over the 56 geometry classes. For inference, an orthographic projection of a real architectural fragment—specifically, a planar, digitally extracted sub-region from one of four photogrammetric building models—is analyzed. The resulting 14×14 saliency map identifies which spatial patches correspond to each geometry class. These saliency maps are then upsampled to 224×224 and segmented using watershed and Voronoi algorithms—two spatial partitioning techniques representing different philosophies of region construction (topographic basins vs. proximity fields). Each segmented micro-region produces a 56-dimensional feature vector by aggregating patch logits. The features from both synthetic and real fragments are embedded into 2D via UMAP to reveal clustering of real fragment regions with synthetic primitives sharing similar geometric features.

This method does not classify fragments by semantic labels (e.g., “cornice,” “capital”), but instead identifies the closest-matching primitive lineage—the underlying generative operation (e.g., circle→extrude, line→revolve)—for each fragment micro-region. Resemblance is quantified as proximity in the learned feature space: each region is embedded as a 56-dimensional vector of lineage probabilities, and similarity is assessed by distance in this latent space (cf. Chefer et al. 2021; McInnes et al. 2018). This latent-geometry clustering links synthetic geometric primitives and compound projective forms to micro-regions of real, irregular 3D fragment meshes. By shifting from semantic to generative classification, the pipeline enables automated, geometry-driven cataloging and analysis of architectural fragments.

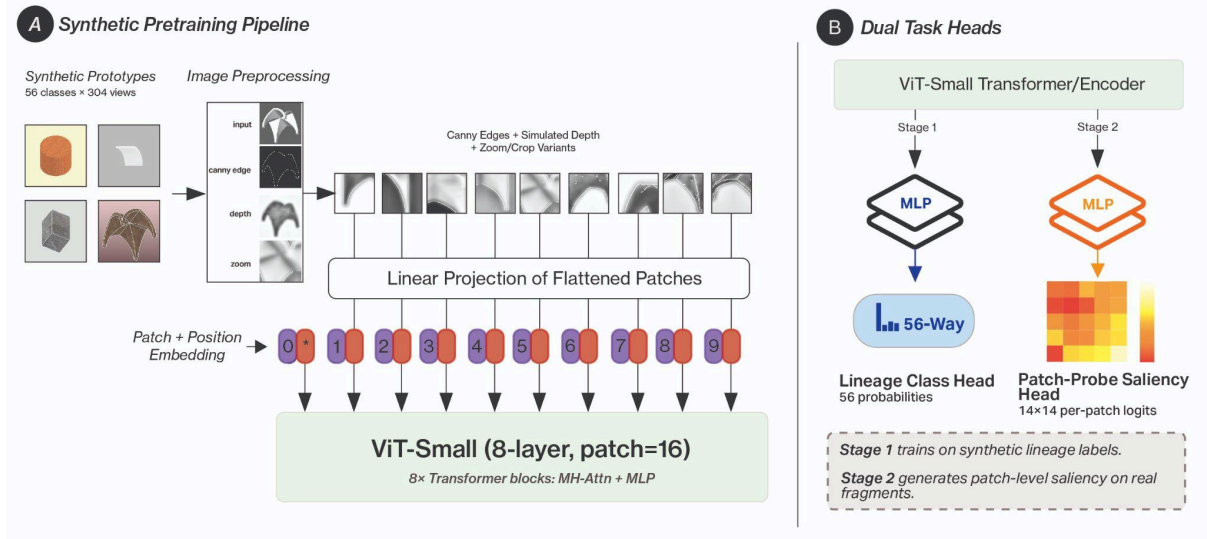


Figure 1. Two-stage architecture overview. (A) Synthetic pretraining pipeline: high-resolution Rhino renders are preprocessed and split into patch embeddings, which feed into a Vision Transformer (ViT-Small/16). (B) Task heads: the frozen ViT produces (1) a global lineage classification head (56 classes) and (2) a patch-probe saliency head (14×14 per-patch logits) for fragment analysis.

This paper’s key contributions are:

1. A two-stage ViT + patch-probe pipeline that (i) pretrains on synthetic lineage labels; (ii) computes saliency maps on real fragment renders to reveal latent projective cues.
2. A multi-scale segmentation ($\sigma = 2, 5, 10$) of upsampled 14×14 saliency into discrete micro-regions, each represented by a 56-dim “lineage-prob” vector.
3. A UMAP embedding in which synthetic primitives and fragment micro-regions co-cluster according to shared geometry (e.g., “cone truncations,” “cylinder cuts”), demonstrating cross-domain generalization.

The remainder of the paper is organized as follows. Section 2 reviews related work in fragment analysis, projective geometry, and transformer saliency. Section 3 details our Synthetic Dataset (Rhino primitives) and our Fragment Dataset (photogrammetry-extracted orthographies). Section 4 describes the Model Architecture and the two training stages (ViT pretraining + patch-probe). Section 5 explains the Saliency + Segmentation pipeline and how region-features are computed. Section 6 presents UMAP embeddings and quantitative/qualitative results showing how fragments map to synthetic primitives. Section 7 discusses implications for architectural reconstruction and outlines limitations. Section 8 concludes and describes future work—specifically, how to extend this pipeline to full 3D fragments or to quantify “drift” across building phases.

2. Related Work

2.1. Computational Fragment Analysis

Classical architectural fragment studies (Choisy 1873; Evans 1995) rely on manual drafting to reconstruct “latent geometry” (e.g., bay widths, vault curvatures). More recent digital efforts (Dore & Murphy 2013; Zheliazkova et al. 2015) apply 3D scanning and point-cloud

segmentation to recover fragment surfaces, but these typically require expensive hardware and human guidance to fit parametric surfaces. Our approach, by contrast, works from orthographic 2D renders of fragments—captured from photogrammetry—and uses AI to recover the same projective cues without explicit 3D reconstruction.

2.2. Projective Geometry in Vision

Inferring conic sections or projective transformations from 2D imagery has a long tradition (Faugeras 1993; Hartley & Zisserman 2003). In architecture, Dore & Murphy (2013) use parametric modeling to analyze façade geometry, while Duarte et al. (2019) employ CNNs for damage detection on building façades. However, these methods still rely on direct geometry fitting or semantic segmentation. In contrast, Vision Transformers (ViTs) (Dosovitskiy et al. 2021) naturally process an image as a sequence of patch tokens, enabling fine-grained localization of projective features without gradient-based saliency. Recent work (Chefer et al. 2021; Liu et al. 2021) shows that a small "patch-probe" head appended to a frozen ViT can yield 14×14 patch-level probability grids for arbitrary labels. We build on this to recover projective geometry from synthetic and real domains.

2.3. Saliency Segmentation & Embedding

While numerous techniques exist for segmenting saliency maps, the key challenge lies in selecting segmentation methods that best align with the geometric goals of the analysis. Watershed segmentation (Vincent & Soille 1991) interprets the saliency map as a topographic surface, delineating regions by local basins, while Voronoi tessellation (Okabe et al. 2000) partitions space by proximity to salient peaks. Both have established use in saliency-based object proposals (Goferman et al. 2012). In this work, both approaches are applied at multiple smoothing scales ($\sigma = 2, 5, 10$) to isolate micro-regions corresponding to distinct projective features. Each segmented region's 56-dimensional patch-probability vector is then embedded into 2D space via UMAP (McInnes et al. 2018) to reveal latent geometric clusters.

Figures and tables should be centered, with one per line. Captions should be in 10 pt text and directly above tables and directly below the figures. Examples are shown below.

3. Datasets

2.1. Class Definitions and Generative Lineages

Our synthetic dataset comprises 56 geometry classes, each produced in Rhino via a fixed, reproducible "recipe" of primitive and constructive operations. We organize these classes into six morphological families:

- Vaults (e.g. Rib Vault, Full; Groin Vault, Half; Gothic Barrel Vault, Quarter)
- Domes (e.g. Dome, Quarter; Gothic Dome, Quarter)
- Spherical Segments (Sphere; Hemisphere; Quarter Sphere; Eighth Sphere)

- Cones & Cylinders (Cone; Truncated Cone; Cylinder; Half Cylinder; Compressed Cylinder)
- Prisms & Cuboids (Cube; Rectangular Cuboid; Slender Rectangular Cuboid; Variable Rectangular Prism)
- Pyramids & Arches (4-Sided Pyramid; 5-Sided Truncated Pyramid; Semicircular Arch; Segmental Arch, Square Base; Ogee Arch; Catenary Arch)

Each class label in our dataset matches Figure 3 (e.g., “Rib Vault, Full,” “Segmental Arch, Half”). Geometries are generated from a 2D primitive (Circle, Line, or Polygon) using one or two operations: Extrude, Revolve, or BooleanDifference.

Example lineage recipes:

- CIRCLE → EXTRUDE → Cylinder
- CIRCLE → EXTRUDE → BOOLDIF_RECTANGLE → Half Cylinder
- CIRCLE → REVOLVE → Torus (Donut)
- POLYGON_TRI → EXTRUDE → Triangular Prism

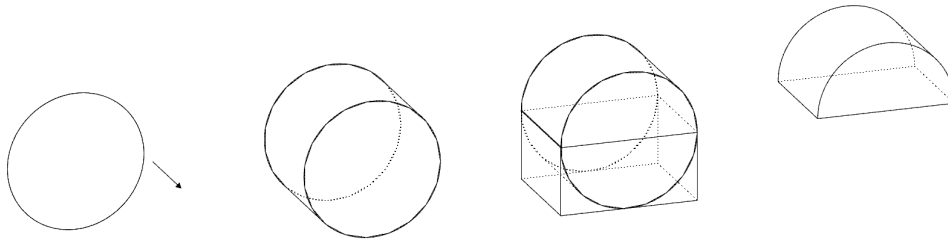


Figure 2 : *Sample Generative Taxonomy: “CIRCLE → EXTRUDE → BOOLSUB_RECTANGLE” → Final Class = Half Cylinder*

3.1.2 Rendering Protocol

For each of the 56 geometry classes, we used a Python-based rendering pipeline in Rhino to ensure consistent and comprehensive coverage of geometric variation. The process began by instantiating a base primitive (for example, a circle of 1 m radius), to which we applied a fixed sequence of operations—such as extrusion, Boolean difference, or revolution—to produce the target geometry. To minimize visual bias, all models were assigned a uniform matte-gray material. Each object was then rendered from 38 unique isometric camera positions, covering a range of horizontal rotations and elevation angles, including both top and bottom views (with view symmetry considered to avoid redundancy). For every camera angle, we rendered eight variations under different combinations of lighting, ground plane, and material presets, resulting in a comprehensive set of appearance conditions. Each configuration was exported as a 224×224 PNG image, already center-cropped and edge-padded in Rhino to standardize the dataset. In total, this protocol produced 304 images per class and 17,024 images overall, which were divided into training (≈70%), validation (≈20%), and testing (≈10%) sets.



Figure 3: 56 classes, samples

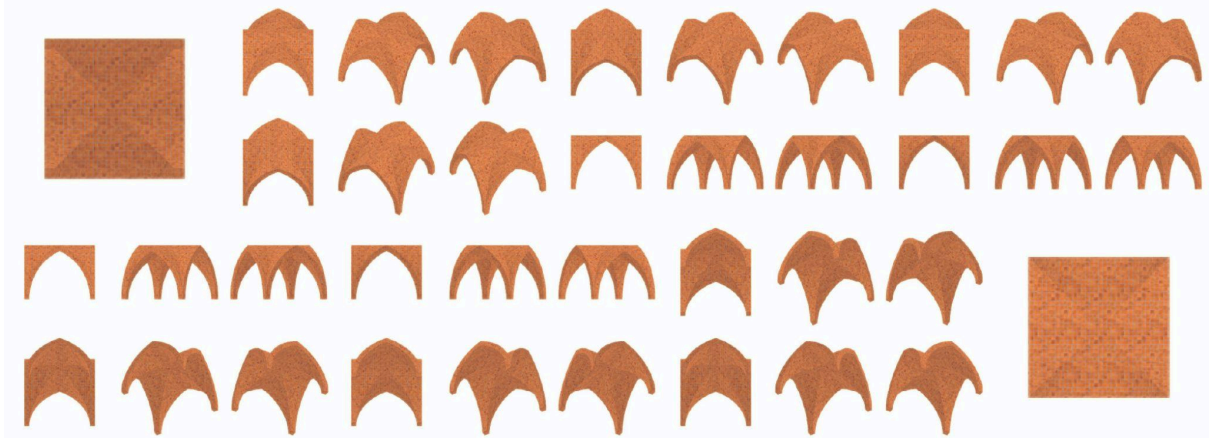


Figure 4: All isometric-orientation renders of the Gothic rib-vault primitive used in our synthetic dataset, showing every unique horizontal rotation (azimuth) and elevation. Views are sampled on a sphere about the model: two poles ($\phi = 0^\circ, 180^\circ$) and three mid-latitude rings ($\phi = 45^\circ, 90^\circ, 135^\circ$). Each ring is captured at $\Delta\theta = 30^\circ$ azimuth increments ($\theta = 0^\circ \dots 330^\circ$), yielding 38 distinct frames in total ($2 + 3 \times 12$). The camera targets the bounding-box center; the radius equals the bounding-box diagonal $\times$ *RadiusFactor*. Larger thumbnails denote orientations that are symmetrically equivalent under the vault’s four-fold plan symmetry, where one image represents four azimuths.

3.2 Real Photogrammetric Fragments

3.2.1 Source Buildings & Fragment Extraction

From four historic buildings (Buildings A–D), we acquired high-resolution photogrammetry models (dense point clouds meshed into OBJ). These buildings include:

- **Building A:** Taxiarches Monastery in Thessaloniki, Greece (*model: Haroupas, 2021*).
- **Building B:** Holy Monastery of Hosios Loukas in Hosios Loukas, Greece (*model: Archéomatique, 2019*).
- **Building C:** La Chapelle Saint-Yves in Rennes, France (*model: Archéomatique, 2018*).
- **Building D:** Chiesa Madre Santa Maria la Nova in Chiaramonte Gulfi, Italy (*model: filarosa, 2021*).

Each photogrammetric model enabled the generation of orthographic (parallel-projection) renders—measured views devoid of perspective distortion and directly analogous to idealized projections in projective geometry, not to be confused with photographic images. For each building, orthographic PNGs (2048×2048) were produced from multiple elevations, from which 20–25 fragment regions (e.g., vault ribs, capital corners, cornice slices) were manually clipped as polygons. Each fragment was exported as a 512×512 PNG with transparent background, then downsampled to 224×224 with edge-padding to preserve geometric fidelity and aspect ratio.

In total, we collected **83 fragments** (≈ 20 per building). We reserve 75% for “fragment inference” and hold out 25% as “fragment test” for final evaluation.

3.2.2 Preprocessing & Orthographic Consistency

Because fragments are rendered orthographically from 3D models, all images represent true parallel projections of measured surfaces—ensuring that geometric analysis is free from the distortions inherent in perspective photography. Fragments were centered and padded to 224×224 pixels without further augmentation (such as flipping or rotation), preserving their original orientation. This decision was made to respect the architectural significance of orientation and to observe model responses to authentic geometric context.

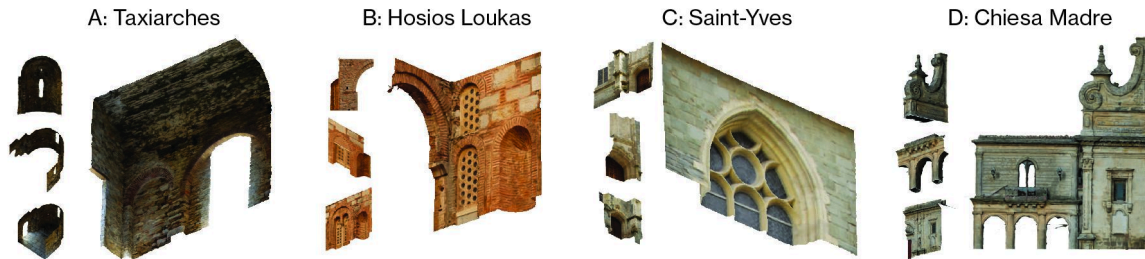


Figure 5: Sample Fragment Clips from Building A-D, showcasing stylistic diversity in test set. Orthographies derive from the following Sketchfab models: Haroupas (2021); Archéomatique (2018, 2019); filarosa (2021).

4. Model Architecture & Training

4.1 Pipeline Introduction

The proposed analysis pipeline comprises two main stages: (1) pretraining a vision transformer (ViT) to classify synthetic geometry classes from rendered images, and (2) adapting this backbone with a specialized “patch-probe” head for saliency-based fragment analysis. This integrated architecture is designed to first capture the latent projective transformations embedded in synthetic shapes, and then transfer that learned geometric sensitivity to the analysis of real architectural fragments.

4.2 Stage 1: ViT Pre Training on Synthetic Lineage Classes

The backbone of our system is a Vision Transformer (ViT-Small/16), initially pretrained on ImageNet-1k. This architecture was selected for its ability to encode complex spatial relationships from image patches, making it especially suitable for identifying subtle geometric features and projective operations within rendered architectural forms.

To adapt the ViT to our domain-specific task of classifying 56 synthetic geometry classes, we replaced the standard classification head with a custom two-layer neural network. The new head expands the ViT’s 384-dimensional feature vectors to 512 dimensions (using a ReLU activation and dropout to minimize overfitting), and then projects to a 56-dimensional output—each dimension corresponding to a unique generative lineage.

Training was conducted using the AdamW optimizer, with distinct learning rates: a lower rate ($1e-4$) for the pretrained backbone (to prevent catastrophic forgetting of spatial representations), and a higher rate ($3e-4$) for the new classification head (to enable rapid

adaptation to the new task). This approach follows established best practices for fine-tuning large pretrained models. A cosine decay schedule with a linear warm-up over 30 epochs was applied to ensure stable and efficient convergence.

To enhance robustness and maintain consistency across evaluation images, we applied comprehensive data augmentation—Canny edge overlays, random resized crops, flips, and rotations—to the training, validation, and test splits. These augmentations reduce model sensitivity to trivial variations and help ensure consistent performance across diverse geometric perspectives. All images were normalized to a pixel range of $[-1, 1]$. The network’s performance was monitored on a dedicated validation set, with a **final top-1 classification accuracy of 54.19%**. While modest, this accuracy is significant given the task’s complexity—accurately discriminating subtle geometric transformations across numerous visually similar classes—and demonstrates sufficient learning of latent geometric features essential for meaningful transfer to real-world fragment analysis.

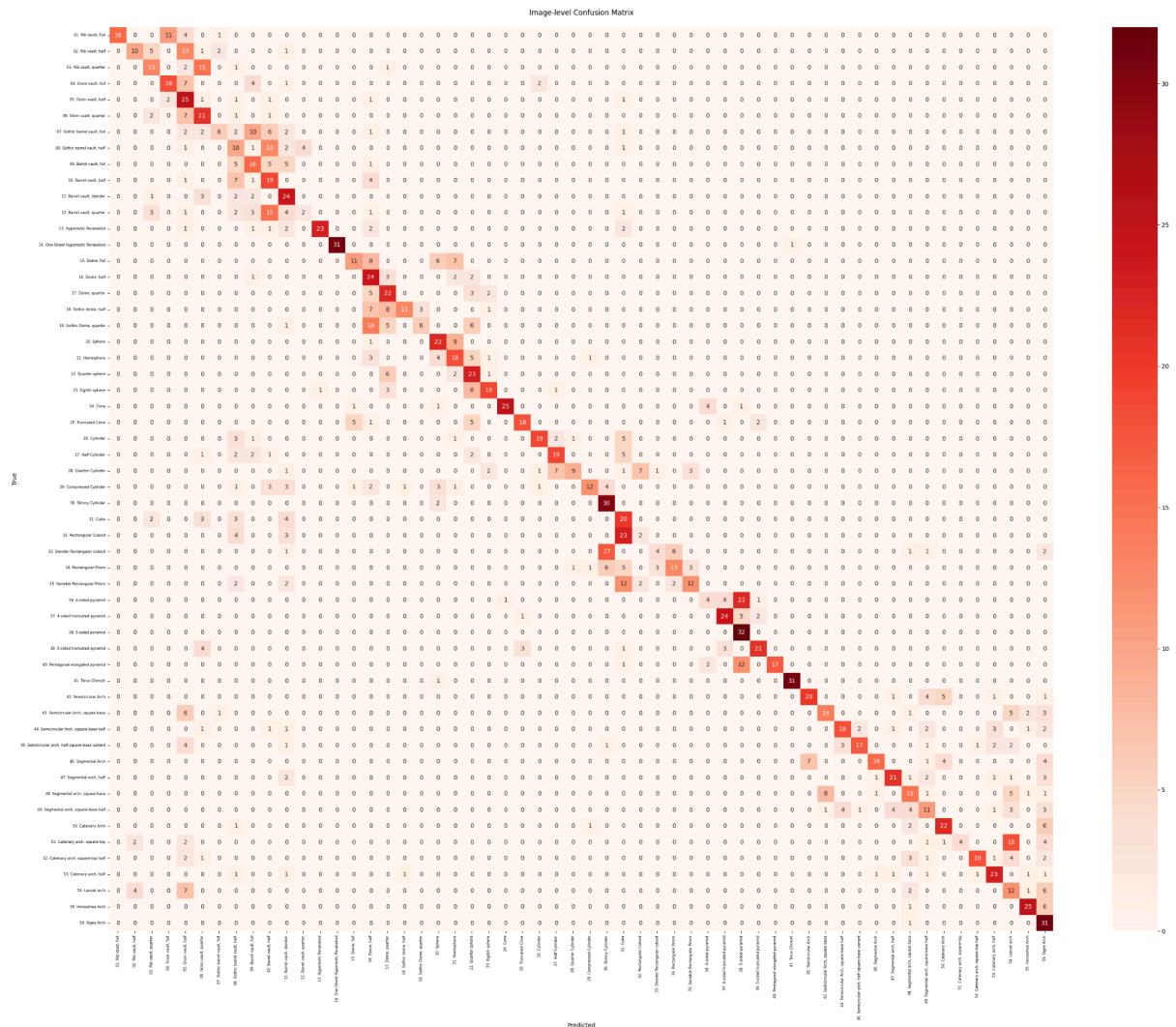


Figure 6: Image-Level Confusion Matrix across 56 classes

4.3 Stage 2: Patch-Probe Saliency & Fragment Inference

Following pretraining, the ViT backbone was frozen to preserve its learned geometric features. A lightweight multilayer perceptron (MLP)—the “patch-probe” head—was appended to the frozen backbone, enabling fine-grained analysis of local image regions (14×14 patches). During training, each patch embedding was labeled according to the geometry class of the full image, allowing the network to specialize in identifying local geometric cues within the broader architectural context.

The patch-probe head was trained with AdamW and a cosine decay learning rate schedule tailored to this new module, promoting effective learning at a fine spatial resolution. The resulting model achieved approximately **48.14% patch-level accuracy**, supporting its ability to localize projective features within complex shapes. Given the fine-grained spatial resolution and inherent ambiguity of localizing projective cues in irregular fragment imagery, this level of accuracy indicates meaningful success in identifying and distinguishing critical geometric characteristics relevant to architectural analysis.

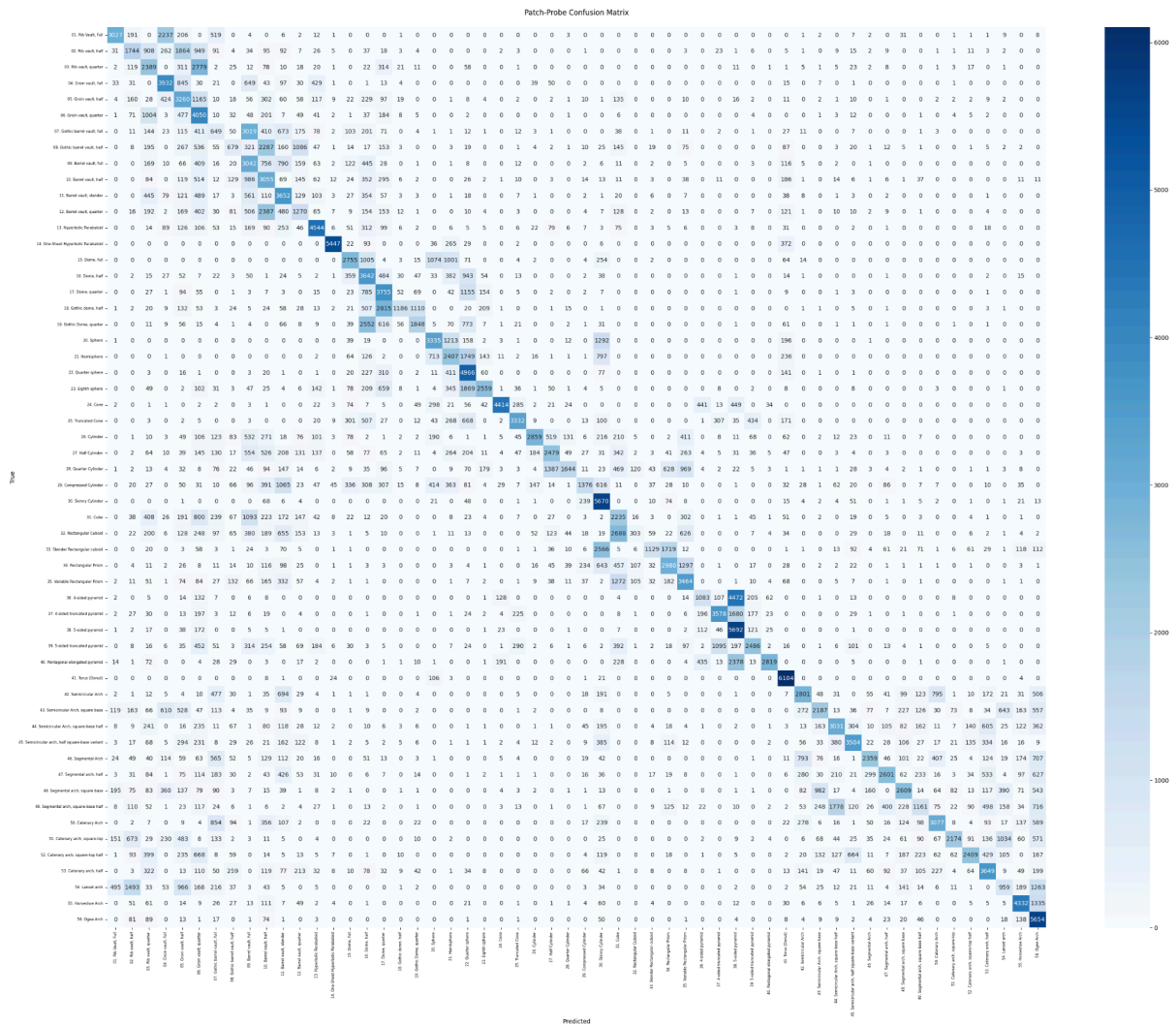


Figure 7: Patch-Probe Confusion Matrix across 56 classes.

4.3.1 Fragment Inference and Saliency Mapping

For inference on real photogrammetric fragments, the model computes patch-wise class probabilities across all geometry classes. Patches with low variance—indicative of background—are masked out, resulting in detailed saliency maps that highlight the regions of each fragment most closely associated with learned geometric classes. These saliency visualizations establish a direct link between the synthetic training regime and the analysis of irregular, real-world architectural fragments.

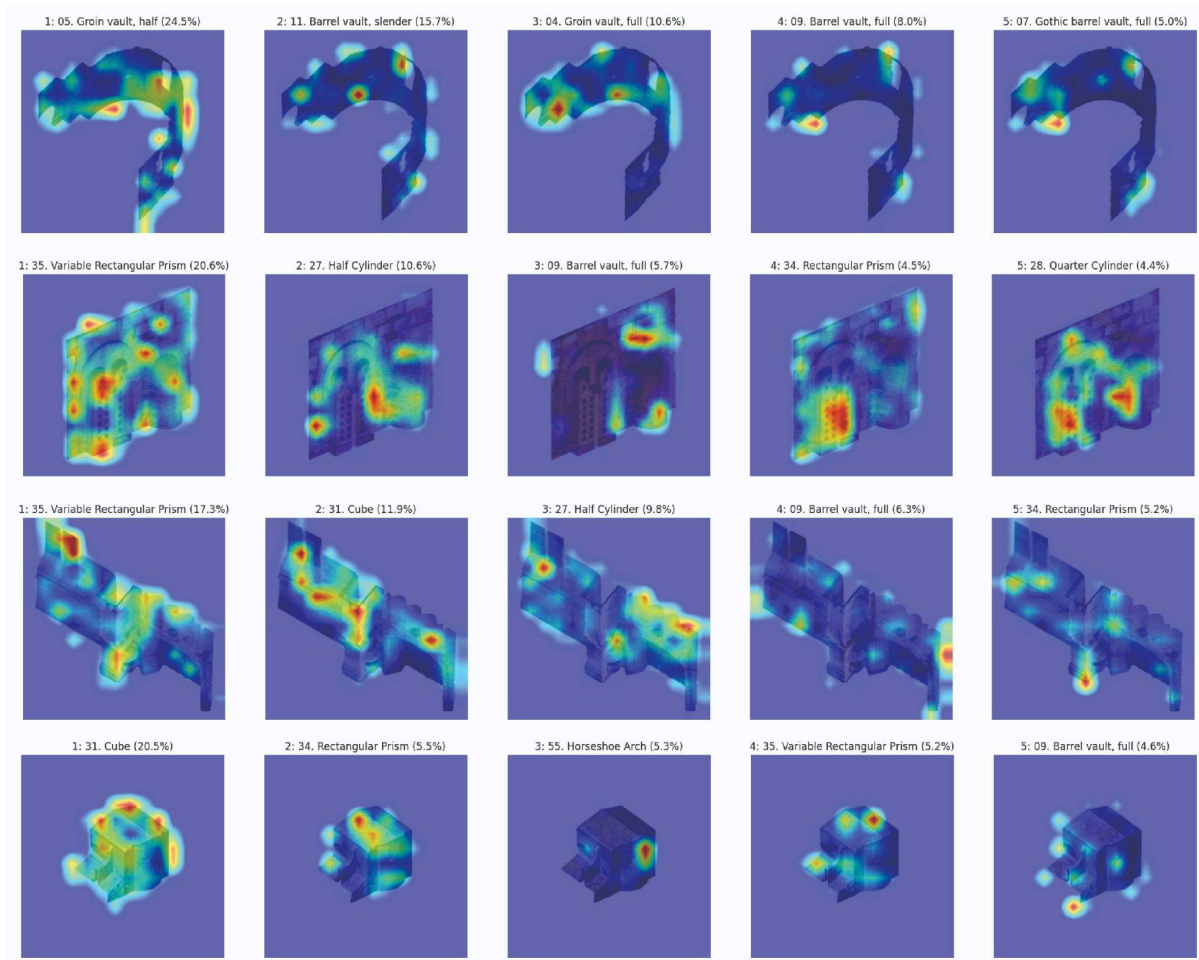


Figure 8: Condensed patch-probe saliency visualization for the top-4 predicted Rhino primitives on exemplar photogrammetric fragments. Each row is a different fragment (from Taxiarches, Hosios Loukas, Saint-Yves, and Chiesa Madre, respectively), and columns 1–5 show the 1st through 5th highest-confidence lineage classes (label + confidence %). Saliency heatmaps are masked to exclude low-variance background patches ($\sigma_p > 0.01$), so only the fragment regions driving each projective-geometry prediction are highlighted.

5. Saliency Segmentation & Region-Feature Extraction

5.1 Multi-Scale Segmentation

Detailed segmentation of saliency maps isolated distinct geometric micro-regions within each fragment. Two complementary techniques—watershed and Voronoi—were utilized across three Gaussian smoothing scales ($\sigma = 2, 5, 10$). Watershed segmentation identified peaks within the saliency landscape to delineate basins, while Voronoi segmentation spatially partitioned the fragments based on the nearest detected saliency peaks.

5.2 Segmentation Evaluation and Selection

Quantitative comparisons between the watershed and Voronoi segmentations at each scale were performed based on average classification confidence of the resulting regions. The segmentation method yielding higher overall confidence was selected, ensuring precision in delineating geometrically significant micro-regions.

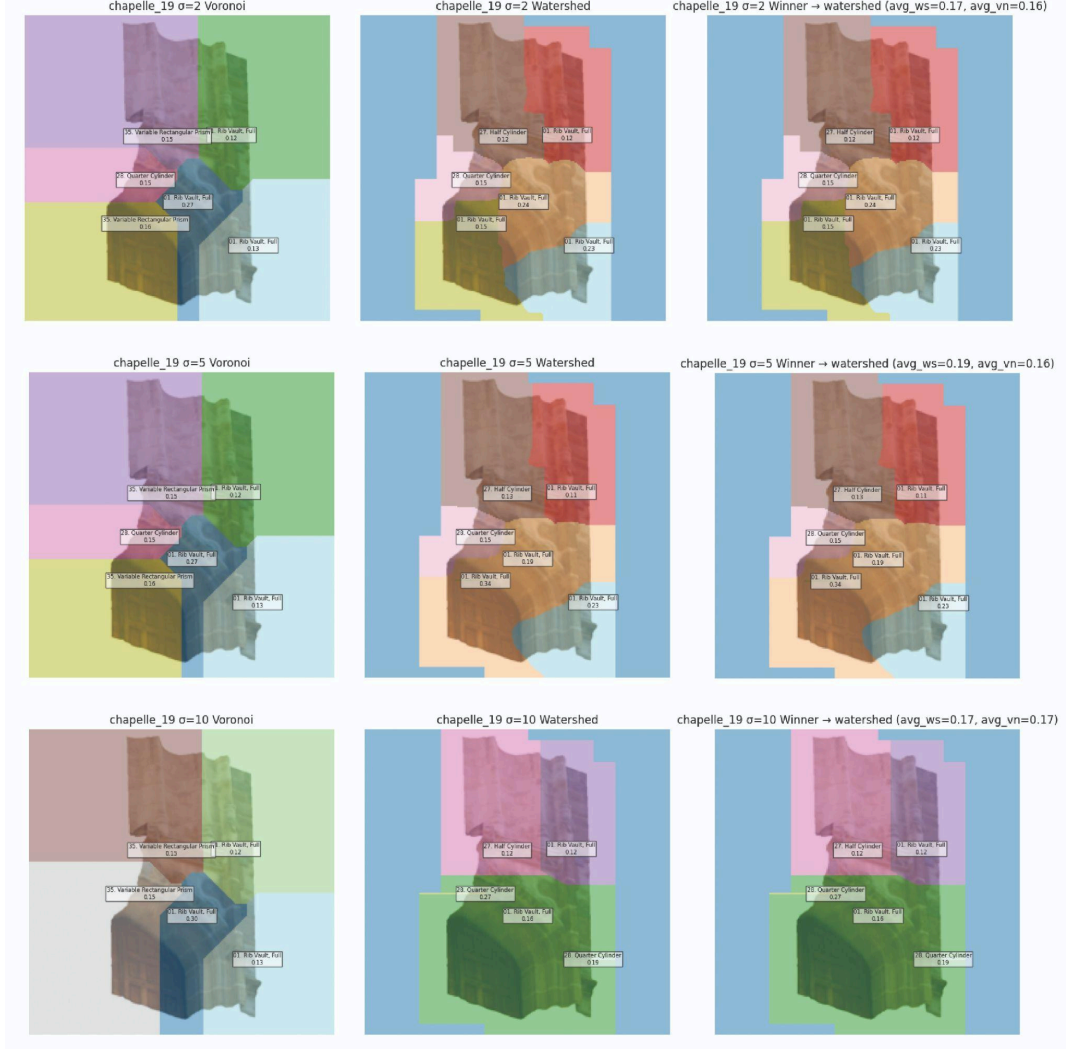


Figure 9: *Multi-scale segmentation and annotation results for fragment chapelle_14*. Each row shows a different Gaussian smoothing scale ($\sigma = 10, 5, 2$). In each row, the left panel is the watershed partition and the middle panel is the Voronoi partition, with regions colored and annotated by their top predicted class and confidence score. The right panel highlights the “winner” segmentation—chosen as whichever method (watershed or Voronoi) has the higher average region confidence (avg_ws vs. avg_vn, reported in the title).

6. Fragment-Level Latent Space Visualization

6.1 Descriptor Aggregation

To enable quantitative comparison across real fragments, we aggregated the patch-level outputs into a single descriptor for each fragment. Specifically, for every segmented micro-region within a fragment (as defined by watershed or Voronoi segmentation), we

computed a 56-dimensional probability vector, corresponding to the model’s confidence across all lineage classes. Each fragment’s descriptor was then obtained by **mean-pooling** these vectors across all its micro-regions. This results in a single 56-dimensional vector that encodes the overall distribution of geometric lineage probabilities for the entire fragment, capturing the relative prevalence of various generative features while preserving underlying geometric relationships.

6.2 UMAP Embedding for Visualization

To interpret these high-dimensional fragment descriptors, we projected them into a two-dimensional latent space using Uniform Manifold Approximation and Projection (UMAP; McInnes et al. 2018). The UMAP was configured with 15 nearest neighbors, a minimum distance of 0.1, and cosine similarity as the distance metric, choices made to emphasize preservation of local relationships and geometric similarity in the embedded space. The resulting 2D coordinates (X, Y) have no inherent geometric units, but the Euclidean distance between points reflects the similarity of fragment geometry as measured by their lineage feature distributions.

Fragments were color-coded by building of origin (Taxiarches Monastery, Hosios Loukas, La Chapelle Saint-Yves, Chiesa Madre Santa Maria la Nova), facilitating direct visual assessment of intra-building consistency and inter-building variation. Clusters in this space indicate groups of fragments with highly similar generative geometry profiles; conversely, separation between points or clusters highlights geometric distinction, either within a building (e.g., stylistic subtypes) or across different buildings and architectural traditions.

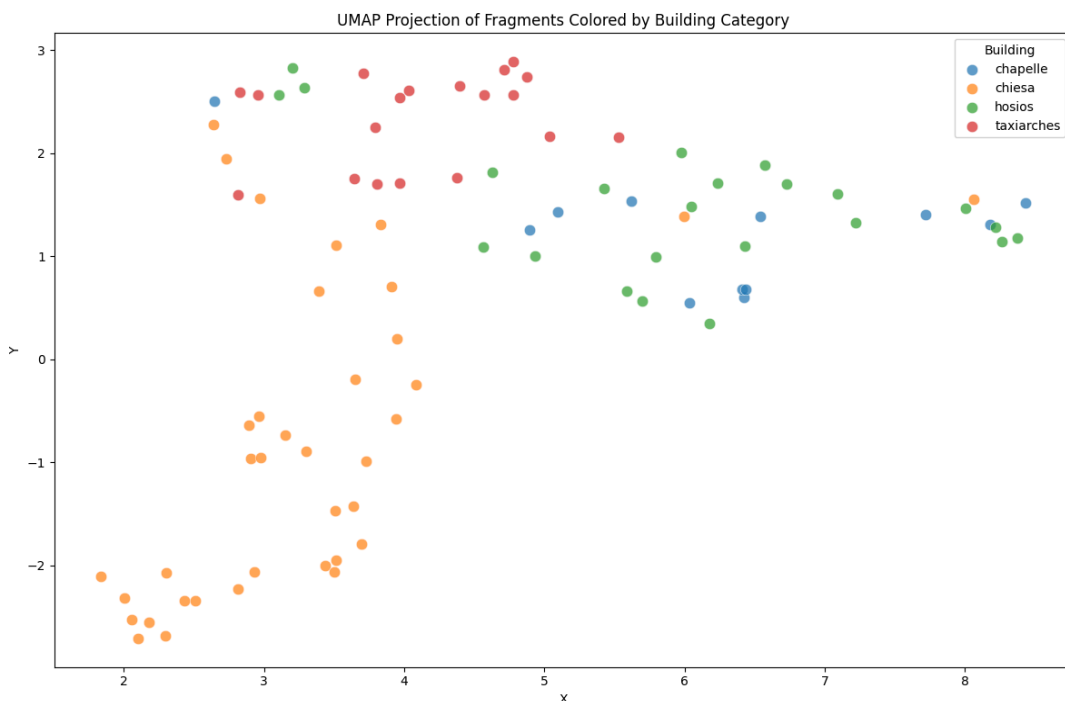


Figure 10. UMAP projection of fragment descriptors (mean-pooled micro-region lineage vectors), with points colored by source building. Clusters represent shared projective geometry, not semantic or typological labels.

6.3 Observations and Interpretations

The UMAP visualization revealed striking patterns: fragments from the same building frequently clustered together, indicating a high degree of internal geometric coherence and suggesting standardized design or construction practices within individual architectural ensembles. At the same time, fragments from different buildings—but with similar geometric lineage signatures—often occupied adjacent regions in the embedded space, pointing to convergent use of fundamental projective geometries (such as truncated cones or cylindrical segments) across geographically and historically distinct sites. This cross-building clustering highlights the enduring role of certain geometric archetypes within architectural history.

Notably, the approach also distinguishes outlier fragments, which either possess unique geometric forms not widely shared, or may reflect errors in segmentation or model inference. These outliers offer a valuable diagnostic for both further method refinement and for identifying particularly distinctive or idiosyncratic architectural solutions.

6.4 Future Enhancements

Looking ahead, several avenues could further strengthen the analytic and practical utility of this framework. Alternative aggregation strategies (e.g., median or weighted pooling) and systematic tuning of UMAP parameters could enhance the robustness of latent embeddings. Integrating additional metadata—such as construction dates, stylistic designations, or geographic context—would enable a richer interpretation of geometric clustering and facilitate longitudinal studies of design evolution.

More broadly, these methods could support interactive visual analytics platforms for architectural historians, practitioners, or regulatory boards. By visually decomposing architectural styles into their underlying projective geometries, stakeholders could quantify the degree of variation within or between architectural projects. Such tools might ultimately inform preservation policy, guide new design by referencing geometric lineage diversity, or even enable objective enforcement of design guidelines (e.g., specifying a minimum “geometric distance” from previous projects to avoid repetitive development patterns). This approach thus offers both new scholarly insight and practical levers for shaping the built environment.

7. Discussion and Implications

7.1 Discussion

This work introduces a two-stage pipeline for analyzing architectural fragments using projective-geometry cues learned from synthetic data. By training a Vision Transformer on rendered lineage classes and transferring this knowledge via a patch-probe mechanism, we demonstrate that meaningful geometric relationships can be recovered from real photogrammetric fragments—even when direct 3D supervision is limited or unavailable. Although the achieved accuracies at the global (54.19%) and patch (48.14%) levels appear moderate, they reflect the inherent complexity and fine-grained nature of the classification tasks. Future enhancements—such as increased dataset diversity (via domain randomization of textures/illumination and light sensor noise, plus mild geometry wear and

occlusions to emulate photogrammetry “distractions”), adaptive segmentation, and higher-resolution or hierarchical inputs—will likely improve accuracy and robustness, further expanding the practical utility and reliability of this methodology in architectural heritage preservation and analysis contexts.

7.2 Implications

The approach enables a shift from purely semantic or typological classification toward objective, geometry-driven mapping of fragments by their generative lineage. Such mapping does not seek to replace conventional taxonomies—such as “cornice” or “vault,” which remain critical to historical and conservation work—but rather supplements them by offering new modes of structural analysis, cross-cultural comparison, preservation decision-making, and the development of design languages rooted in geometric ancestry. The ability to quantify and compare latent projective features opens new avenues for reconstructing lost contexts, understanding architectural innovation, and even entering the generative “mindset” of historical builders.

7.3 Limitations

Several limitations merit emphasis. First, our current Voronoi/marker-watershed pipeline tends to produce cellular regions that negotiate adjacency rather than respecting semantic geometry boundaries (e.g., where a web terminates and a rib begins). A stronger alternative is a boundary- and confidence-aware segmentation that combines (a) boundary probability maps and part-type scores with (b) geometric cues (surface normals, curvature changes, silhouette/crease edges) to constrain the partition. Second, the 224×224 patch granularity may obscure fine architectural detail; a hierarchical scheme that uses smaller patches near high-confidence boundaries and coarser patches in homogeneous regions could recover sharper edges without exploding compute. Finally, the scope of the study remains limited by dataset scale; larger and more diverse sets—paired with weak labels for “geometry groups” (rib, web, pendentive, pier, etc.)—should improve generalization and support the boundary-aware strategy above.

7.4 Future Directions

We advance the projective tradition from interpretive reading to operational inference. Technically, we will enforce cross-view consistency with light 3D cues so that saliency and region descriptors persist under rotation, occlusion, and wear, while keeping compute small. Conceptually, we recast the lineage embedding as an operator over form—a compact, interrogable code for operative geometry that is legible to designers. This reframes fragments as evidence of procedures (ruling, folding, tiling, pendentive construction), enabling retrieval, alignment, and comparison by recoverable action rather than typology. Because the pipeline is patch-explainable and low-compute, we can release a portable corpus, baselines, and an interactive viewer that renders claims reproducible and falsifiable on commodity hardware.

Read architecturally, this yields quantified histories of making: trajectories in operator space (stability, geodesic distance, diachronic drift) test standards and innovations with evidence that links material, structure, light, and view to their generative means. Read computationally, it offers a compact projective index for complex spatial organization — fast enough for search, precise enough for alignment, and interpretable enough for practice. This approach makes projective reasoning practical and portable—bringing latent geometry from model to monument across scholarship, conservation, and design.

References

- Archéomatique. (2018, July 16). *Chapelle St-Yves, Rennes (35)* [3D model]. Sketchfab. <https://sketchfab.com/3d-models/chapelle-st-yves-rennes-35-e6fc5848b36643a285aca762c24422fc> Sketchfab
- Archéomatique. (2019, November 12). *Monastery of Hosios Loukas, Greece* [3D model]. Sketchfab. <https://sketchfab.com/3d-models/monastery-of-hosios-loukas-greece-ffc935ea442443c599aad3cdf9fea3a1>
- Chefer, Hila, Shir Gur, and Lior Wolf. 2021. "Generic Attention-Model Explainability for Interpreting Bi-Models." In Proceedings of the IEEE/CVF International Conference on Computer Vision (ICCV), 9444–9453.
- Choisy, Auguste. 1873. *Histoire de l'architecture*. Paris: Librairie de la Société Française d'Archéologie.
- Dore, Christopher, and Fiona Murphy. 2013. "Integration of Historic Building Information Modeling (HBIM) and 3D GIS for Recording and Managing Cultural Heritage Sites." *ISPRS Annals of the Photogrammetry, Remote Sensing and Spatial Information Sciences* 2 (5): 85–92.
- Dosovitskiy, Alexey, et al. 2021. "An Image is Worth 16×16 Words: Transformers for Image Recognition at Scale." In International Conference on Learning Representations (ICLR).
- Duarte, João, Tiago Martins, and João P. Costeira. 2019. "CNN-Based Damage Detection in Building Facades." *Automation in Construction* 106: 102869.
- Evans, Robin. 1995. *The Projective Cast: Architecture and Its Three Geometries*. Cambridge, MA: MIT Press.
- Faugeras, Olivier. 1993. *Three-Dimensional Computer Vision: A Geometric Viewpoint*. Cambridge, MA: MIT Press.
- Fai, Stephen, and Mikael Sydor. 2013. "Parametric Modeling and Surface Reconstruction for Architectural Fragments." In 2013 Digital Heritage International Congress (DigitalHeritage), 731–734. IEEE.
- filarosa. (2021, July 1). *Chiesa Madre Santa Maria la Nova* [3D model]. Sketchfab. <https://sketchfab.com/3d-models/chiesa-madre-santa-maria-la-nova-30e9c77e302043fbb37fc34c5fd6ef32>
- Goferman, Stas, Lihi Zelnik-Manor, and Ayellet Tal. 2012. "Context-Aware Saliency Detection." *IEEE Transactions on Pattern Analysis and Machine Intelligence* 34 (10): 1915–1926.
- Haroupas, V. (2021, February 3). *Taxiarches Interior Test Scan* [3D model]. Sketchfab. <https://sketchfab.com/3d-models/taxiarches-interior-test-scan-4e2f9e1ed5f14995ab7963a24dca1678> Sketchfab

- Hartley, Richard, and Andrew Zisserman. 2003. *Multiple View Geometry in Computer Vision*. 2nd ed. Cambridge: Cambridge University Press.
- Liu, Zhaowei, et al. 2021. "Visual Saliency Transformer." In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 3665–3674.
- McInnes, Leland, John Healy, and James Melville. 2018. "UMAP: Uniform Manifold Approximation and Projection for Dimension Reduction." *Journal of Open Source Software* 3 (29): 861.
- Okabe, Atsuyuki, Barry Boots, Kokichi Sugihara, and Sung Nok Chiu. 2000. *Spatial Tessellations: Concepts and Applications of Voronoi Diagrams*. 2nd ed. New York: Wiley.
- Radford, Alec, Jong Wook Kim, Chris Hallacy, Aditya Ramesh, Gabriel Goh, Sandhini Agarwal, Girish Sastry, et al. 2021. "Learning Transferable Visual Models From Natural Language Supervision." OpenAI. <https://arxiv.org/abs/2103.00020>.
- Vincent, Luc, and Pierre Soille. 1991. "Watersheds in Digital Spaces: An Efficient Algorithm Based on Immersion Simulations." *IEEE Transactions on Pattern Analysis and Machine Intelligence* 13 (6): 583–598.
- Zheliazkova, Nadezhda, et al. 2015. "Parametric Reconstruction of Architectural Heritage from Point Clouds." *ISPRS Annals of Photogrammetry, Remote Sensing and Spatial Information Sciences* II-5/W3: 169–176.